

Overview of AlphaFold

Subject: AlphaFold

1.

AlphaFold 1, 2, and AlphaFold DB AlphaFold DB provides models of individual protein chains (monomers), rather than their biologically relevant complexes. Many protein regions are predicted with low confidence score, including the intrinsically disordered protein regions. AlphaFold-2 was validated for predicting effects of point mutations on structure and free energy, with a partial success.

2.

Reception AlphaFold 2 scoring more than 90 in CASP's global distance test (GDT) is a great achievement in computational biology. Nobel Prize winner and structural biologist Venki Ramakrishnan called the result "a stunning advance on the protein folding problem", adding that "It has occurred decades before many people in the field would have predicted. It will be exciting to see the many ways in which it will fundamentally change biological research." AlphaFold 2's success received wide media attention. News pieces appeared in the science press, such as Nature, Science, MIT Technology Review, and New Scientist, and the story was covered by national newspapers. A frequent theme was the ability to predict protein structures based on the constituent amino acid sequence, expected to have benefits in the life sciences—accelerating drug discovery and enabling better understanding of diseases. Some have noted that even a perfect answer to the protein prediction problem still leaves questions about the protein folding problem (and thus protein dynamics)—understanding in detail how the folding process actually occurs in nature (and how sometimes they can also misfold). In 2023, Demis Hassabis and John Jumper won the Breakthrough Prize in Life Sciences as well as the Albert Lasker Award for Basic Medical Research for their management of the AlphaFold project. Hassabis and Jumper proceeded to win the Nobel Prize in Chemistry in 2024 for their work on "protein structure prediction" with David Baker of the University of Washington.

3.

Announced on 8 May 2024, AlphaFold 3 was co-developed by Google DeepMind and Isomorphic Labs, both subsidiaries of Alphabet. AlphaFold 3 is not limited to proteins, as it can also predict the structures of protein complexes with DNA, RNA, post-translational modifications and selected ligands and ions. AlphaFold 3 introduces the "Pairformer," a deep learning architecture inspired by the transformer, which is considered similar to, but simpler than, the Evoformer used in AlphaFold 2. The Pairformer module's initial predictions are refined by a diffusion model. This model begins with a cloud of atoms and iteratively refines their positions, guided by the Pairformer's output, to generate a 3D representation of the molecular structure. The AlphaFold server was created to provide free access to AlphaFold 3 for non-commercial research. As of November 2025, the AlphaFold 3 research paper has been directly cited more than 9,000 times.

4.

The AlphaFold Protein Structure Database (AFDB), a joint project between AlphaFold and EMBL-EBI, was launched on July 22, 2021. At launch, the database contained AlphaFold 1-predicted

models for nearly the complete UniProt proteome of humans and 20 model organisms, totaling over 365,000 proteins. The database does not include proteins with fewer than 16 or more than 2700 amino acid residues, but for humans they are available in the whole batch file. AlphaFold's initial goal (as of early 2022) was to expand the database to cover most of the UniRef90 set, which contains over 100 million proteins. As of May 15, 2022, the database contained 992,316 predictions. In July 2021, UniProt-KB and InterPro has been updated to show AlphaFold predictions when available. On July 28, 2022, the team uploaded to the database the structures of around 200 million proteins from 1 million species, covering nearly every known protein on the planet. The number as of 2024 is 214 million, with 26 million being duplicates (exact sequence matches) of another protein in the database. The predicted structures can differ significantly between duplicates. As of 2025, the AFDB uses AlphaFold 2 for its predictions. All structures produced remain monomeric, but multimeric structures produced by other databases are linked on the page through the 3D-Beacons API. Foldseek, which provides fast and accurate structure searches, is also integrated. Information from AlphaMissense (a tool that uses AlphaFold to predict the outcome of missense mutations) is also integrated.

5.

Published works Andrew W. Senior et al. (December 2019), "Protein structure prediction using multiple deep neural networks in the 13th Critical Assessment of Protein Structure Prediction (CASP13)", *Proteins: Structure, Function, Bioinformatics* 87(12) 1141–1148 doi:10.1002/prot.25834 Andrew W. Senior et al. (15 January 2020), "Improved protein structure prediction using potentials from deep learning", *Nature* 577 706–710 doi:10.1038/s41586-019-1923-7 John Jumper et al. (December 2020), "High Accuracy Protein Structure Prediction Using Deep Learning", in *Fourteenth Critical Assessment of Techniques for Protein Structure Prediction (Abstract Book)*, pp. 22–24 John Jumper et al. (December 2020), "AlphaFold 2". Presentation given at CASP 14. Abramson, J., Adler, J., Dunger, J. et al. (May 2024), "Accurate structure prediction of biomolecular interactions with AlphaFold 3", *Nature* 630, 493–500 (2024)

6.

Further reading Carlos Outeiral, CASP14: what Google DeepMind's AlphaFold 2 really achieved, and what it means for protein folding, biology and bioinformatics, Oxford Protein Informatics Group. (3 December) Mohammed AlQuraishi, AlphaFold2 @ CASP14: "It feels like one's child has left home." (blog), 8 December 2020 Mohammed AlQuraishi, The AlphaFold2 Method Paper: A Fount of Good Ideas (blog), 25 July 2021

7.

AlphaFold has been used to predict structures of proteins of SARS-CoV-2, the causative agent of COVID-19. The structures of these proteins were pending experimental detection in early 2020. Results were reviewed by scientists at the Francis Crick Institute in the United Kingdom before being released to the broader research community. The team also confirmed accurate prediction against the experimentally determined SARS-CoV-2 spike protein that was shared in the Protein Data Bank, an international open-access database, before releasing the computationally determined structures of the under-studied protein molecules. The team acknowledged that although these protein structures might not be the subject of ongoing therapeutical research efforts, they will add to the community's understanding of the SARS-CoV-2 virus. Specifically, AlphaFold 2's prediction of the structure of the ORF3a protein was very similar to the structure determined by researchers at University of California,

Berkeley using cryo-electron microscopy. This specific protein is believed to assist the virus in breaking out of the host cell once it replicates. This protein is also believed to play a role in triggering the inflammatory response to the infection.